

ET AI HACKATHON 2026 · PHASE 2 · PROBLEM #2 — ENERGY SUPPLY-CHAIN
RESILIENCE

KARNADHAR

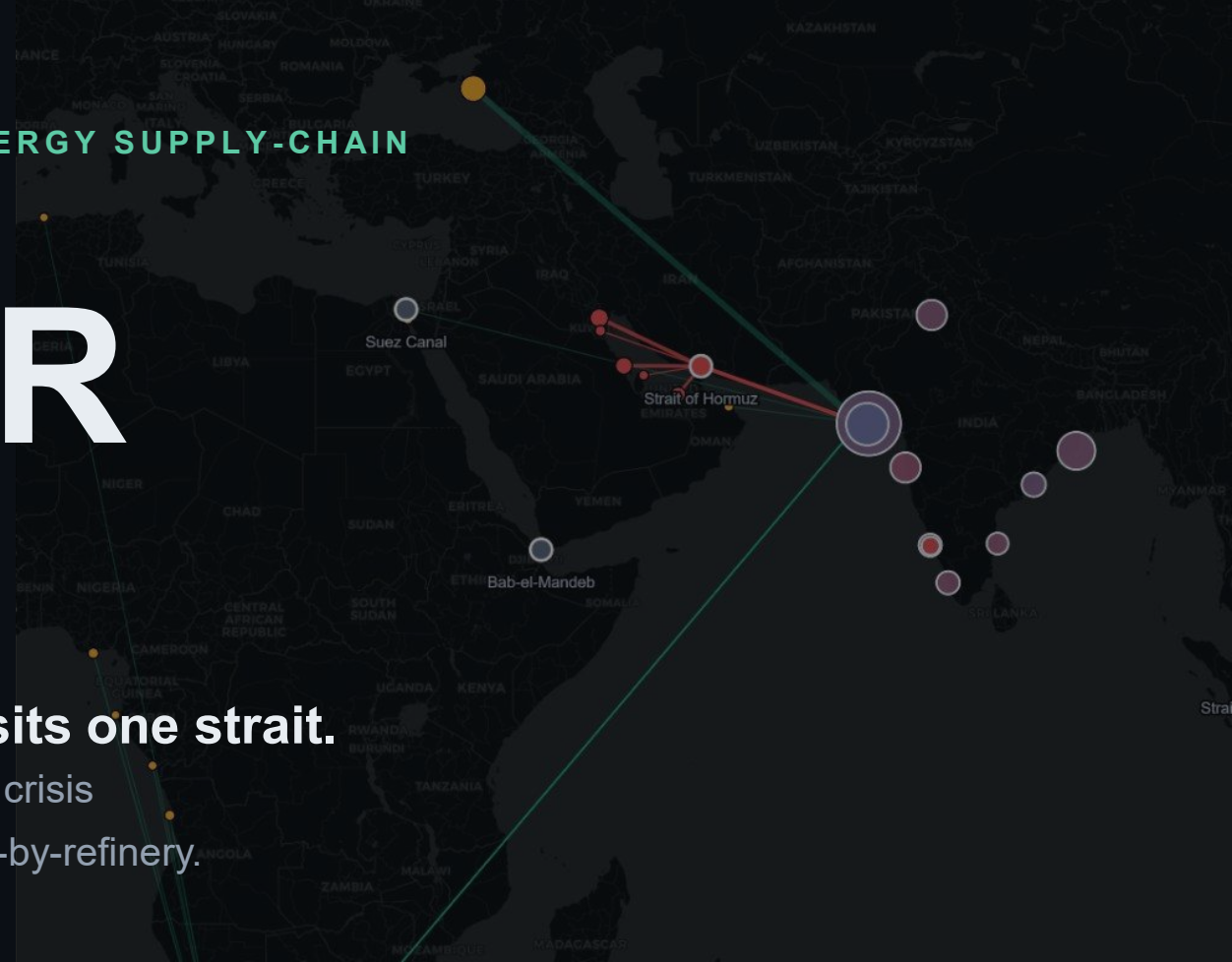
कर्णधार — the helmsman who steers the ship through the storm

India imports 88% of its crude. Nearly half transits one strait.

KARNADHAR is the AI command center that turns a 47-day supply crisis into a decision made in 45 milliseconds — grade-by-grade, refinery-by-refinery.

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Assumptions reviewed with: Lydia Powell (ORF) · Prof. U. K. Bhui (PDEU)



THE PROBLEM

One strait can strangle the world's 3rd-largest oil importer

88%

of India's crude is imported

46%

transits the Strait of Hormuz —
derived from official trade records,
not assumed

9.5 days

of strategic reserve cover —
voluntary, vs the IEA's 90-day norm

+8%

Brent in a single session when
Hormuz was threatened (June 2025)

And the real constraint isn't barrels — it's dollars.

India is a twin-deficit economy. A sustained price shock adds ~\$190 bn/yr to the import bill, blowing the current-account deficit from 1.2% to over 6% of GDP. Unlike trade-surplus Japan or Korea, India cannot sustain that dollar outflow — which is why unprepared economies take **47 days longer** to stabilise supply (McKinsey). The data to act exists. The intelligence layer to act on it does not.

Crude oil is not fungible

Every other “resilience dashboard” treats oil as one number: barrels.

Reality: a refinery is an engine tuned to a fuel. A deep-conversion coking refinery built for heavy, sour, asphaltene-rich crude physically cannot run ultra-light condensate — and a simple refinery cannot digest sour crude. When Hormuz closes, the question is not “where do we buy oil?” It is “which exact grade can each of our refineries actually run — and who has it?”

“That is — you are perfectly right.”

Prof. U. K. Bhui, Petroleum Engineering, PDEU — on our refinery-constrained sourcing thesis. His guidance added the SARA dimension (asphaltenes) to our crude model.

WHAT DECIDES IF A BARREL IS RUNNABLE

- **API gravity** — light vs heavy
- **Sulphur** — sweet vs sour
- **Asphaltene (SARA)** — coke precursor

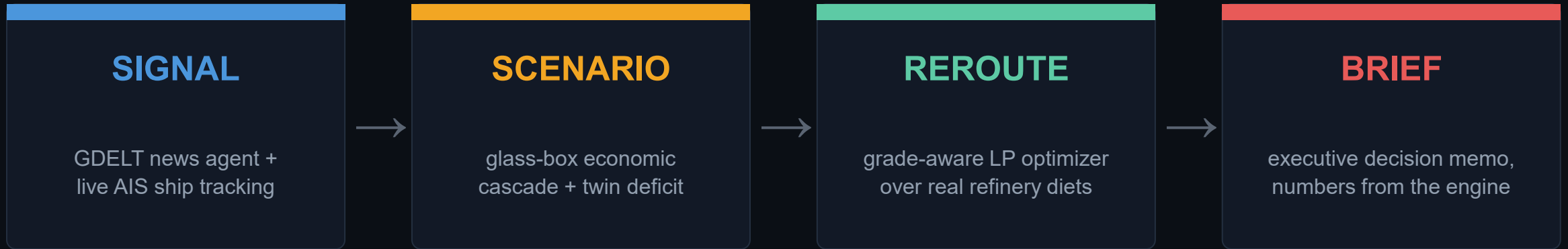
SAME COUNTRY, DIFFERENT MACHINES

RIL Jamnagar **Nelson 21.1**
runs almost anything — needs heavy feed

BPCL/HPCL Mumbai **Nelson 7.0**
light, sweeter diet only — sour is a hard limit

WHAT WE BUILT

A multi-agent brain: signal → scenario → reroute → brief



- REAL public data — no synthetic signals
- Explicit, testable assumptions (every parameter named & editable)
- Knowledge graph: supplier → route → chokepoint → refinery (50 nodes, 163 edges)
- LLM writes words, never numbers — engine stays auditable

45 ms

end-to-end: signal to executable recommendation

vs the 47-day average for economies without response intelligence

Built on official trade records — not synthetic data

Refinery (port)	MT/yr	Nelson	Hormuz exposure	Top real suppliers
RIL Jamnagar (Sikka + SEZ)	76.3	21.1	43%	Russia 35 · Saudi 17 · Iraq 14
Nayara Vadinar	41.7	11.8	29%	Russia 56 · Iraq 18 · Angola 6
IOCL Paradip	34.8	12.2	50%	Iraq 24 · Russia 21 · UAE 14
HMEL Bathinda (Mundra)	23.3	12.5	52%	Russia 32 · Saudi 18 · Iraq 16
BPCL/HPCL Mumbai	23.2	7.0	60%	Saudi 25 · UAE 23 · US 21
BPCL Kochi	12.4	8.0	54%	Russia 31 · UAE 23 · Iraq 16

46%

national Hormuz exposure —
DERIVED from customs data

10 × 36

refineries × supplier grades,
real volumes & landed prices

Every refinery’s crude diet, Hormuz dependence, and even its processing limits are grounded in what it actually imported — DGCIS port-wise records, PPAC capacities, public Nelson complexity, crude assay libraries.

The “fungible” plan is physically impossible. Ours is not.

NAIVE · “oil is oil”

INFEASIBLE

250 kb/d of ultra-light condensate assigned to deep coking refineries that **physically cannot run it**. A grade failure — not a logistics failure. This is the plan a fungible model would execute.

KARNADHAR · grade-aware LP

FEASIBLE

Every barrel runnable, sourced, within desulphurisation and asphaltene capacity. Heavy-sour rationed to the cokers (Jamnagar, Paradip); light-sweet routed to simple refineries (Mumbai, Chennai). **Executable by a procurement desk.**

\$3.15 M/day

product yield protected by grade-matching

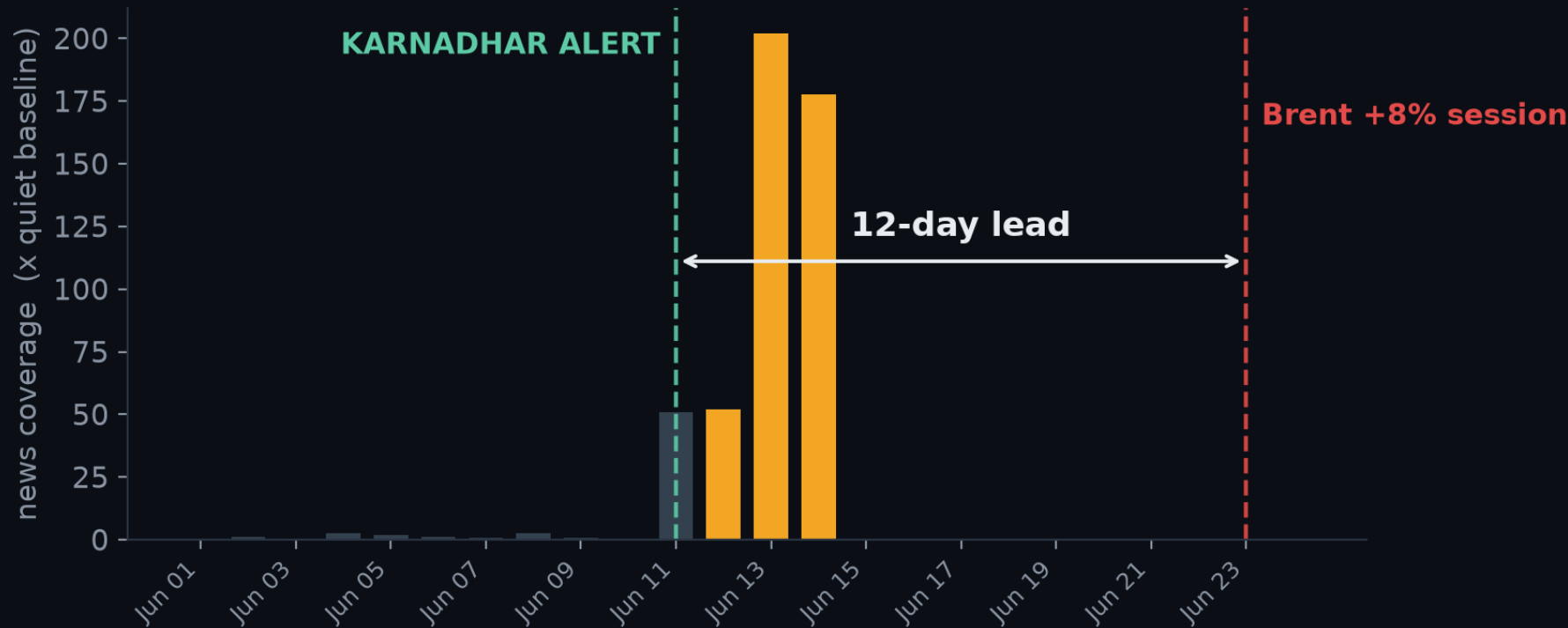
~\$1.15 bn/yr

annualised value of that decision quality

1,370 kb/d

re-sourced across 10 refineries in one solve

We saw the June 2025 Hormuz crisis 12 days before the market



12 days

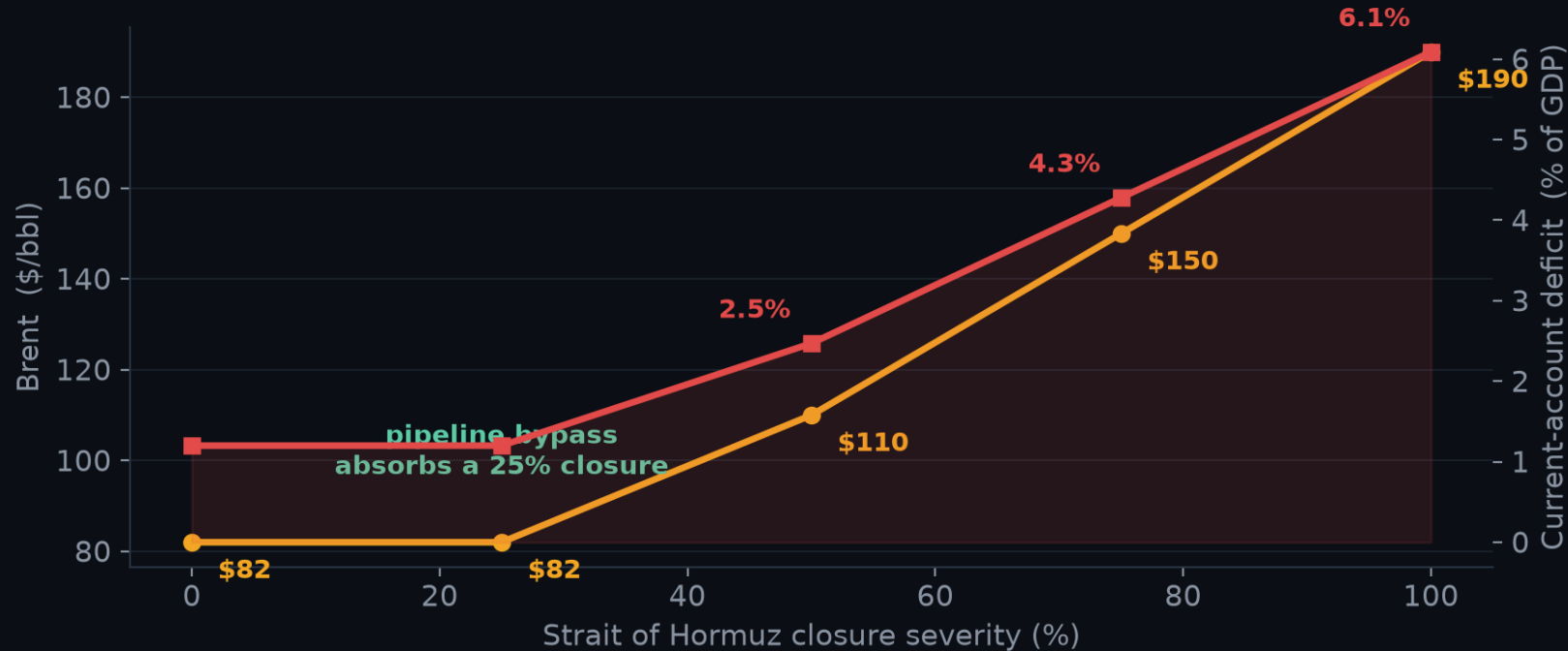
alert lead over the
Brent +8% session

200×

peak news coverage vs
quiet baseline (real GDELТ)

Method: 10-day rolling baseline → alert when coverage sustains $\geq 3\times$ baseline for 2 days. Simple, explainable, testable — and fused with live AIS tanker traffic (congestion flags at chokepoints) for multi-source confirmation.

A glass-box cascade — ending in the deficit India cannot pay



₹125.5/L

retail fuel at full closure (+₹25.5)

-1.4 pp

GDP drag · +2.0 pp CPI

\$191 bn/yr

extra import bill in USD

1.2% → 6.1%

current-account deficit, % of GDP

Every parameter is named, sourced and editable; sensitivity is one function call. “The binding constraint is the balance of payments, not the barrels” — the twin-deficit reframe came from our review with ORF.

Block any chokepoint. Sanction any supplier. Model it.

Hormuz full closure

naive: INFEASIBLE — blend breaches Vizag's sulphur limit

KARNADHAR: FEASIBLE — and priced: the reroute ties up +66 VLCC-equivalents (ton-mile)

Hormuz + OPEC+ squeeze

alternate suppliers squeezed to 60% of headroom

still feasible — grade-matching value rises to \$1.95M/day of protected yield

Russia sanctioned

naive sends 220 kb/d to refineries that cannot run it

usable shortfall 220 → 0 — India's #1 supplier is replaceable, at a price

Hormuz + Russia (compound)

the nightmare: 3,900 kb/d gap, 2 of top 3 suppliers gone

honest: 2,166 kb/d unmet, quantified — naive pretends, with 546 kb/d of fake fill

Also modelled: the brief's Red Sea suspension — India's crude barely moves (Cape-routed), but the commodity lens shows who is hit: Black-Sea edible oils (18%). And when a shock is too severe to reroute around, KARNADHAR quantifies the gap — the SPR scheduler answers hold / bridge / ration with a drawdown rate.

The war-room: India's energy lifeline, live on one screen



- Reroute deep-dive — routes ranked #1..N**
click a disruption: alternative corridors ranked by LP-committed volume, each self-defending; click one to isolate it on the map with its derivation
- Optimizer flows — the plan, drawn**
every LP allocation becomes an animated arc: source → corridor → the exact refinery, width = kb/d
- 3D globe / 2D toggle · interactive KG**
globe or flat map, real AIS vessels, bypass pipelines; zoom/pan/drag the 50-node supplier→strait→refinery graph
- Multi-commodity lens**
same disruption scored across 8 strategic imports — Hormuz hits 53% of LNG too
- Decision layer**
shadow prices, VLCC ton-mile, SPR drawdown schedule + executive brief — on every scenario

WHY YOU CAN TRUST IT

Validated, evaluated, and reviewed by domain experts

AUTOMATED VALIDATION

57 / 57

checks passed — one command, exit-code gated, run in CI on every push
wedge holds: naive infeasible for grade reasons, LP optimal

- ✓ cascade monotonic; bypass nuance; sensitivity band; twin deficit
- ✓ signal alert precedes market repricing on real data
- ✓ real-diet integrity: 46% exposure, configs sane
- ✓ engine admits unmet demand; SPR scheduler holds/bridges/rations
- ✓ decision layer proven: shadow prices · VLCC ton-mile · usable shortfall
- ✓ reroute routes ranked by LP-committed volume — not the scarcity-artifact trap
- ✓ multi-commodity lens generalises — Red Sea spares crude, hits edible oils

EVALUATION FOCUS — ALL FIVE, COVERED

- EF-1 signal lead time: 12 days, real GDELT + AIS
- EF-2 procurement executability: feasible, grade-matched plans
- EF-3 scenario fidelity: explicit, testable, sensitivity-swept
- EF-4 geospatial depth: live routed map, real diets
- EF-5 response time: 45 ms signal → recommendation

EXPERT REVIEW

Lydia Powell — Distinguished Fellow, ORF: twin-deficit framing; SPR-as-voluntary-insurance; sourcing-first policy sequence.

Prof. U. K. Bhui — Petroleum Engg., PDEU: wedge validated (“you are perfectly right”); SARA/asphaltene dimension added on his guidance.

BUSINESS IMPACT & SCALABILITY

Incumbents tell you what’s happening. We tell you what to do.

Kpler, Vortexa, S&P and Wood Mackenzie built a multi-billion-dollar market selling **monitoring**. KARNADHAR sells the **decision layer** — the feasible, grade-aware action plan — AI-native and India-first.

WHO PAYS

Refiners
IOCL · BPCL · HPCL · RIL · Nayara

Traders
desks living on disruption signals

Shippers & insurers
chokepoint risk = pricing

Government
PPAC · ISPRL · MoPNG — the anchor

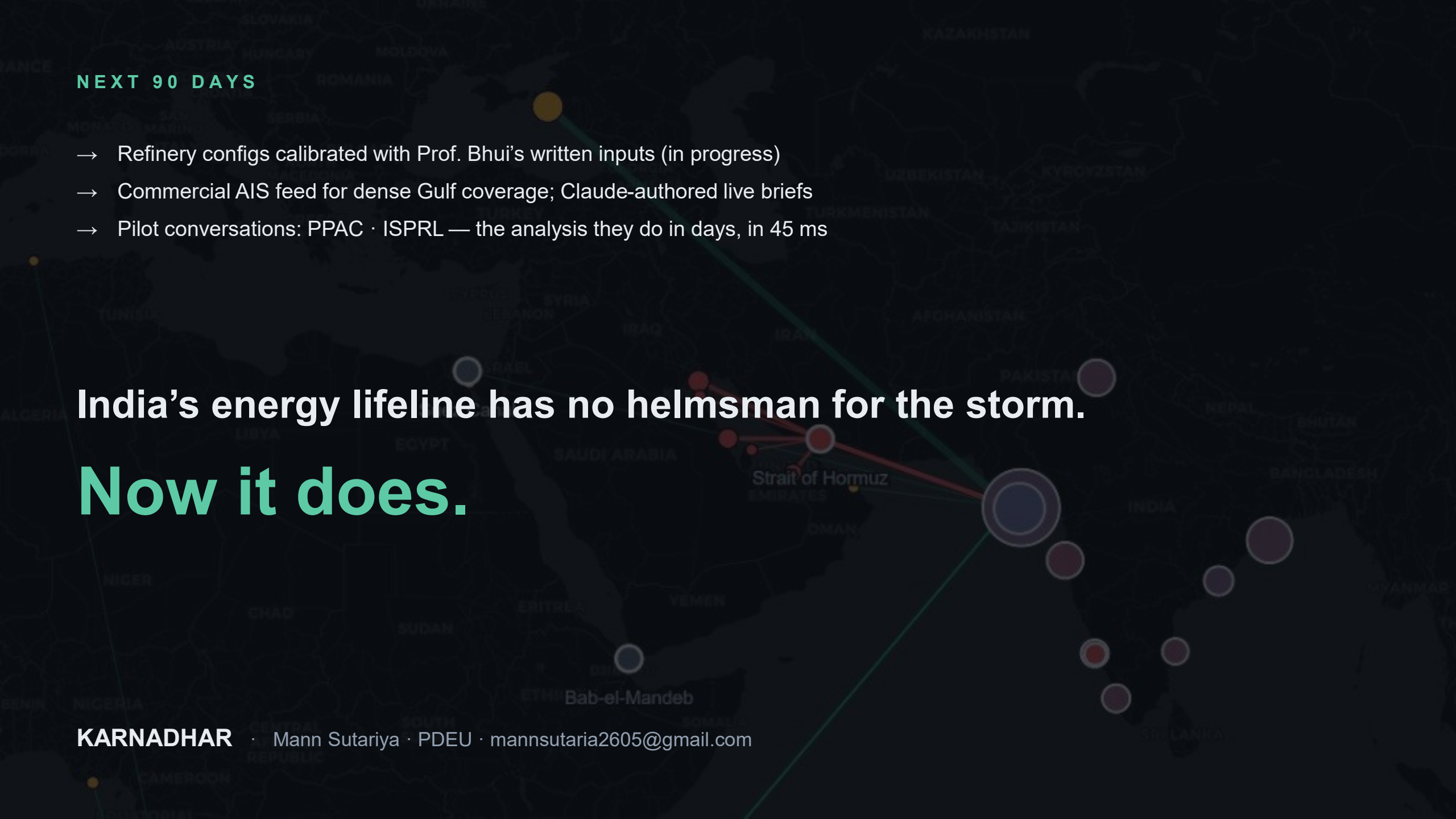
THREE AXES OF SCALE

1 · Segments refiner → trader → insurer → sovereign (same engine, 7 buyer types)

2 · Geography 50+ import-dependent economies share this exact anatomy — Japan, Korea, Turkey, Philippines...

3 · Commodity demonstrated LIVE in-app: 8 strategic imports screened — LNG, pharma APIs, semiconductors... (Hormuz hits 53% of LNG)

Model: tiered SaaS — Monitor → Simulate → Orchestrate — plus API licensing. Land commercial desks first (months, not years); the sovereign deployment is the moat and the mission.

A dark-themed map of the Middle East and surrounding regions, including parts of Europe, North Africa, and South Asia. The map highlights several key locations and infrastructure. A green line runs from a point in the Black Sea (near Ukraine/Russia) through the Strait of Hormuz and down the eastern coast of India. Another green line runs from a point in the Red Sea (near Bab-el-Mandeb) up towards the Strait of Hormuz. Various colored circles (yellow, blue, red, purple) are placed at strategic locations: a yellow circle in the Black Sea, a blue circle in the Red Sea near Bab-el-Mandeb, a red circle in the Persian Gulf near the Strait of Hormuz, and several purple circles along the Indian coast and in the Indian Ocean. The text 'NEXT 90 DAYS' is in green, while the main title and list are in white. The background map shows country borders and names in a light grey font.

NEXT 90 DAYS

- Refinery configs calibrated with Prof. Bhui's written inputs (in progress)
- Commercial AIS feed for dense Gulf coverage; Claude-authored live briefs
- Pilot conversations: PPAC · ISPRL — the analysis they do in days, in 45 ms

India's energy lifeline has no helmsman for the storm.
Now it does.